

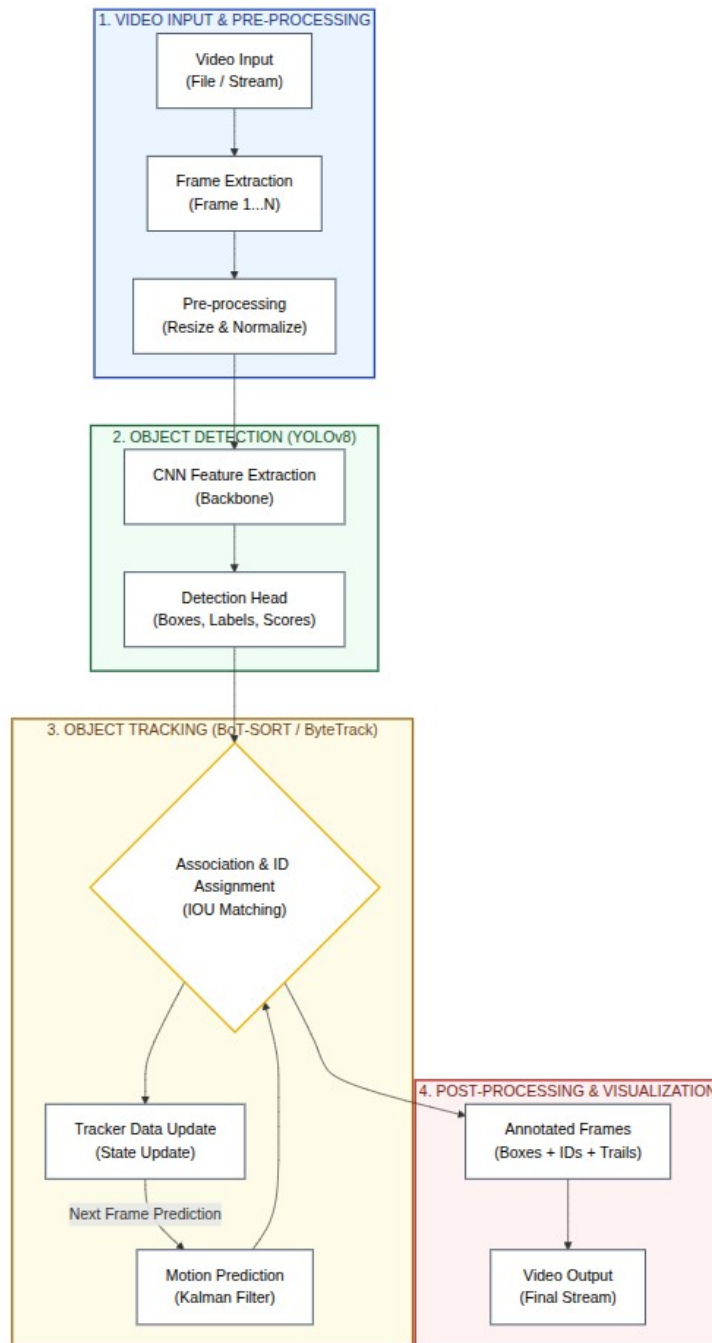
Edge Intelligence

Lab 9

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25MML0043

Design and implement the pipeline for object detection in Videos using YOLOv8

Architecture



Code

1. Install dependencies

```
!pip install ultralytics
```

```
import cv2
```

```
from ultralytics import YOLO
```

```
from google.colab.patches import cv2_imshow
```

```
from IPython.display import clear_output
```

2. Load the YOLOv8 model

```
model = YOLO("yolov8n.pt")
```

3. Path to your video (Ensure you upload the file to Colab first)

```
video_path = "/content/026c7465-309fbd33.mp4"
```

```
cap = cv2.VideoCapture(video_path)
```

Loop through the video frames

```
while cap.isOpened():
```

```
    success, frame = cap.read()
```

```
    if success:
```

```
        # 4. Run YOLOv8 tracking (same logic as your code)
```

```
        # results is a list, so we take results[0]
```

```
        results = model.track(frame, persist=True, tracker="botsort.yaml")
```

```
        # 5. Visualize the results
```

```
        annotated_frame = results[0].plot()
```

```
        # 6. Colab-specific display logic
```

```
        clear_output(wait=True) # Clears previous frame for smooth display
```

```
        cv2_imshow(annotated_frame)
```

```
    else:
```

```
        break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

Output

